Decentralized Edge AI Environmental Hazard Network: First Prototype Stage (PS) Architecture and Feasibility

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***Abstract*—Traditional meteorological monitoring relies on centralized, high-cost macro-stations that report generalized weather data over broad geographic radii. This paradigm degrades sharply in mountainous topographies, where elevation change, ridge shadowing, and valley confinement create sharply isolated microclimates. We present the First Prototype Stage (PS) of a Decentralized Edge AI Environmental Hazard Network: a two-node pair of standalone, solar-powered Edge AI nodes, each built around the ESP32-S3, a 12-parameter environmental and hazard sensor suite, an on-device boosted-tree inference pipeline, and a LoRa (865 MHz) multi-hop mesh radio layer, enabling first-order validation of multi-hop routing between nodes. Each node performs sensor fusion and anomaly classification locally, transmitting only event-driven, low-bandwidth alerts rather than raw telemetry, and remains fully operational during cellular and internet blackouts. We detail the compute, sensing, intelligence, and communication layers of the architecture, contrast the design against commercial monitoring stations, and present a bill-of-materials-based cost analysis, repriced to July 2026 market rates, showing a per-node cost of approximately INR 19,130 (full hazard configuration) and a two-node PS build cost of approximately INR 50,000 including shared NRE, calibration jigs, spare sensors, and contingency, against INR 1,20,000–2,50,000 per unit for comparable commercial stations. We conclude with a scaling and optimization roadmap toward a custom 4-layer PCB and mesh-wide deployment.**

***Index Terms*—Edge AI, LoRa mesh network, environmental sensing, microclimate monitoring, disaster early-warning, ESP32-S3, TinyML, wildfire detection, flash flood prediction**

I. Introduction

Traditional meteorological monitoring relies on centralized, high-cost macro-stations that provide generalized weather data for a broad geographic radius. While effective for flat terrain, this paradigm fundamentally fails in mountainous topographies characterized by complex elevation changes, steep ridges, and deep valleys. The unique topography surrounding the IIT Mandi campus and the greater Himachal Pradesh region creates isolated microclimates, where weather on one side of a ridge can drastically differ from the other.

The First Prototype Stage (PS) of the Edge AI Portable Weather Station project aims to validate a paradigm shift: moving from centralized data collection to a decentralized, self-healing swarm of hyper-local intelligence nodes. The objective of the PS is to design, fabricate, and test a pair of standalone Edge AI nodes that integrate high-fidelity atmospheric and environmental hazard sensors, validating both single-node sensing performance and inter-node LoRa mesh handoff.

By executing machine learning models directly on the microcontroller (Edge AI), the node locally processes raw sensor telemetry, identifies patterns indicative of imminent environmental hazards such as flash floods or approaching wildfire smoke, and transmits low-bandwidth alerts via a LoRa mesh network. The PS serves as the fundamental building block: once the standalone node's hardware, firmware, and power budget are validated, the architecture can scale into a spatial mesh network for real-time tracking of localized weather phenomena without reliance on vulnerable cellular infrastructure.

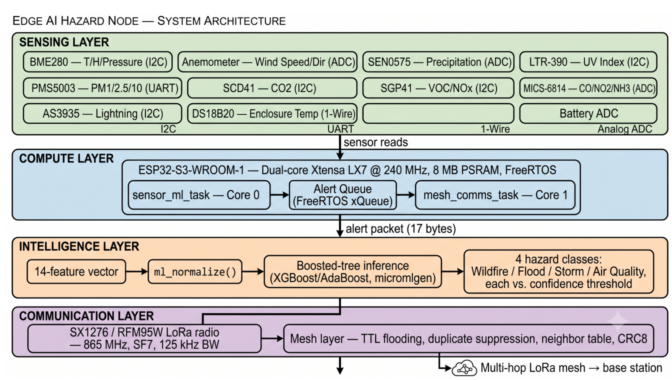

Fig. 1: High-level system architecture of a single Edge AI hazard node, showing the Compute, Sensing, Intelligence, and Communication layers.

II. Approach and System Architecture

Developing the PS requires a multi-disciplinary approach synthesizing custom PCB design, ultra-low-power embedded systems, environmental physics, and machine learning. To avoid compounding errors, the architecture is isolated into four distinct layers: Compute, Sensing, Intelligence, and Communication, as shown in Fig. 1.

A. The Compute Core: ESP32-S3

The processing engine of the node is the ESP32-S3-WROOM-1 module. Selected for its integrated Processor Instruction Extensions (PIE), the S3 provides hardware-accelerated 128-bit vector math, critical for executing neural networks and decision trees efficiently on battery power. The S3 offers 45 programmable GPIO pins, allowing simultaneous operation of an SPI bus (for LoRa), an I2C bus (for multiple environmental sensors), hardware UART (for optical particulate counters), and external hardware interrupts, all buffered by 8 MB of PSRAM to handle dynamic mesh routing tables.

B. The Sensor Array

The PS is equipped with a robust sensor suite monitoring 12 distinct parameters, categorized into baseline weather and severe environmental hazards (Table I).

- **Core Meteorological Baseline:** Temperature, humidity, and barometric pressure via the Bosch BME280 (I2C). Wind speed and direction via a generic combo anemometer and wind vane. Precipitation via a piezoelectric acoustic sensor (DFRobot SEN0575), avoiding the leveling requirement of tipping-bucket designs. UV Index via the LTR390.
- **Hazard and Air Quality Array:** A PMS5003 laser-scattering sensor (UART) for PM2.5/PM10 particulate counting. A Sensirion SCD41 photoacoustic NDIR sensor for true CO_2. An SGP41 for VOCs and NOx. A MICS-6814 for multi-gas detection (CO, NO2, NH3).
- **Predictive Hardware:** An AS3935 lightning sensor analyzes RF emissions to estimate distance to approaching storm fronts up to 40 km away, providing a leading indicator for the predictive models.
- **System Telemetry:** On-board 18650 battery voltage and internal enclosure temperature (DS18B20) for power-cycle regulation and water-ingress detection.

TABLE I: Sensor Array Summary

Category	**Component**	**Interface**
Temp/Hum/Pressure	Bosch BME280	I2C
Wind Speed/Dir	Generic Anemometer + Wind Vane	Analog/Digital
Precipitation	DFRobot SEN0575 (piezo)	Analog
UV Index	LTR390	I2C
PM2.5 / PM10	Plantower PMS5003	UART
CO_2	Sensirion SCD41	I2C
VOC / NOx	Sensirion SGP41	I2C
Multi-gas (CO, NO2, NH3)	MICS-6814	Analog
Lightning distance	AS3935	I2C/SPI
Enclosure temp	DS18B20	1-Wire
Battery voltage	Onboard ADC divider	Analog

C. Edge AI Implementation

Rather than functioning as a passive data logger, the PS acts as an intelligent agent. Using historical weather data, regression models based on boosting algorithms (XGBoost or AdaBoost) are trained offline in Python. These models use feature combinations, e.g., rapid barometric drops coupled with rising PM2.5 levels and decreasing lightning distances, to classify and predict localized anomalies.

Using conversion toolkits such as `micromlgen`, the trained model is exported as a raw C++ header and integrated into the ESP32-S3 firmware. The Ultra-Low-Power (ULP) coprocessor continuously polls the sensors in the background; only when specific thresholds are breached does the main dual-core processor wake, execute the boosted-tree inference, and determine whether an alert should be broadcast.

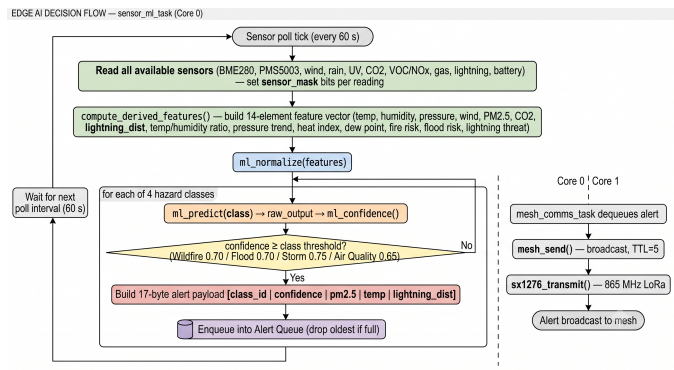

Fig. 2: Edge AI decision flow: from ULP sensor polling to threshold breach, wake, inference, and mesh alert broadcast.

D. Communication Layer: LoRa Mesh

To achieve a 10 km operational range in mountainous terrain without cellular dependency, the PS uses the Semtech SX1276 (RFM95W) transceiver tuned to the legal 865 MHz ISM band. Rather than a direct point-to-point link, which would be blocked by topology, the network employs a multi-hop mesh routing protocol (e.g., Meshtastic or RadioHead). The node transmits event-driven insights, using federated-learning principles to distribute small AI parameter updates or threat alerts to neighboring nodes, eventually routing data to a local base-station dashboard.

E. Enclosure and Power Management

Physical survival of the electronics is provided by an FDM 3D-printed Stevenson screen in PETG filament, ensuring natural airflow while shielding sensors from direct insolation and precipitation. Power is sustained via a high-capacity 18650 lithium-ion cell, managed by a dedicated TP4056 or CN3065 charge controller, and replenished by a 5 W external solar panel.

F. Firmware Implementation

The firmware is implemented in C using the ESP-IDF framework, organized into modular components for sensor management, ML inference, and mesh communication (Fig. 3).

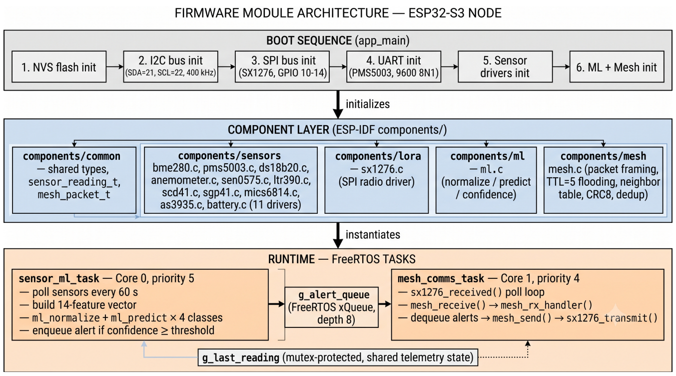

Fig. 3: Firmware architecture: FreeRTOS tasks for sensor reading, ML inference, and mesh communication running on dual-core ESP32-S3.

1) Sensor Pipeline: The sensor pipeline (`sensor_pipeline.c/.h`) provides a Hardware Abstraction Layer (HAL) for all 12 sensors. Each sensor driver handles initialization, calibration, and reading via the appropriate interface:

- **I2C Sensors:** BME280, SCD41, SGP41, LTR390, AS3935
- **UART Sensors:** PMS5003 (particulate matter)
- **1-Wire:** DS18B20 (enclosure temperature)
- **Analog ADC:** MICS-6814 (multi-gas), wind speed/direction

The pipeline computes 14 derived features from raw sensor data, including heat index, dew point, fire risk index, and pressure trends, which are consumed by the ML inference engine.

2) ML Inference Engine: The ML pipeline (`ml_pipeline.c/.h`) implements a complete XGBoost inference engine optimized for ESP32-S3. The model consists of 16 shallow decision trees (max depth 4) per hazard class, processing 14 normalized features through a binary sigmoid output. The inference achieves $\sim$50 μs latency per prediction, enabling real-time hazard detection at 5-second intervals.

3) Mesh Communication: The mesh module (`mesh_comm.c/.h`) handles LoRa communication via the SX1276 transceiver at 865 MHz. It implements multi-hop packet routing with automatic forwarding, heartbeat broadcasting, and model update distribution for federated learning. Packet types include alerts, heartbeats, sensor data, and model updates.

4) Main Entry Point: The main firmware (`main.c`) orchestrates three FreeRTOS tasks:

1) **Sensor Task (Core 0):** Reads sensors, computes features, runs ML inference, generates alerts when thresholds are breached.
2) **Mesh Task (Core 1):** Handles packet transmission, reception, forwarding, and heartbeat broadcasts.
3) **Monitor Task (Core 1):** Periodic status reporting, battery monitoring, and mesh statistics.

III. Difference from Commercial Monitoring Stations

The Edge AI mesh architecture diverges fundamentally from legacy commercial systems, offering distinct operational advantages in constrained, dynamic environments (Table II).

A. Zero-Cloud Survivability

The most significant differentiator is resilience. Commercial IoT stations become inert when cellular networks fail during severe storms. Because the PS runs its predictive algorithms natively on the ESP32-S3 bare metal, it retains full analytical capability offline. The LoRa mesh acts as an independent intranet, bypassing ISP infrastructure to deliver warnings directly to localized physical sirens or base stations.

B. High-Fidelity Hazard Detection

Commercial stations prioritize broad meteorological trends. By integrating the PMS5003 laser particulate counter and Sensirion gas arrays alongside acoustic precipitation and lightning RF detection, the PS doubles as an environmental hazard node, tuned for wildfire smoke tracking and localized flash-flood prediction.

IV. Cost Breakdown and Economic Viability

Commercial weather stations with equivalent granular hazard detection (PM2.5, true CO_2, lightning distance, multi-gas) typically cost between INR 1,20,000 and INR 2,50,000 per unit, rendering dense spatial microclimate mapping financially impractical for independent researchers or local municipalities. By using off-the-shelf breakout modules and the open-source C++ ecosystem, the Edge AI PS achieves comparable analytical density at a fraction of the cost (Table III).

Note: wind speed/direction uses a generic analog anemometer + wind-vane pair (~INR 2,300) rather than the genuine SparkFun Weather Meter Kit, which now retails in India at INR 9,589 due to import markup on branded mechanical sensors — swap in if brand-matched hardware is required.

A. Scalability and Optimization

At approximately INR 19,130 per node, the two-node PS build totals roughly INR 50,000 including shared NRE, calibration jigs, spare sensors, and contingency — itself substantially lower than a single commercial unit (INR 1.2–2.5 Lakhs). Scaling the same per-node rate to a full 20-node mesh network represents a capital expenditure of approximately INR 3.83 Lakhs, still substantially lower than a two-unit commercial deployment. Transitioning from prototype to production mesh, a unified schematic in KiCad and a custom 4-layer PCB integrating all sensor ICs directly (bypassing redundant regulators and LEDs found on commercial breakout boards) is projected to reduce per-node cost by an additional 20–30%. For baseline nodes where hazard gases are not required, removing the SCD41, SGP41, and MICS-6814 reduces cost to approximately INR 13,530, enabling mass spatial deployment

TABLE II: Architectural Comparison Between Legacy and Edge AI Weather Systems

Feature Domain	Traditional Commercial Station	Edge AI Mesh Node
Data Resolution	Macro-level; generalized data over >50 km radius.	Hyper-local; maps precise spatial microclimates and topographical variation.
Processing Paradigm	Passive telemetry; raw, continuous streams to a central cloud server.	Edge inference; ingests raw data locally, runs ML models (e.g., XGBoost), transmits insights.
Connectivity	Fragile; relies on Wi-Fi, 4G, or Satellite IoT; fails during blackouts.	Zero-cloud survivability; ad-hoc LoRa mesh reroutes dynamically around failed nodes.
Bandwidth Usage	High; raw data matrices choke low-bandwidth networks.	Event-driven; small payload alerts only (e.g., "Flood Risk 85%").
Deployment Logistics	Static, heavy; requires professional installation and leveling.	Portable, agile, solar-powered; rapid deployment along supply chains or ridges.

TABLE III: Estimated Bill of Materials for the PS Node Pair (July 2026 pricing)

Category	Components	Per Node	2 Nodes
Compute & Radio	ESP32-S3-WROOM-1 DevKit N16R8, RFM95W SX1276 (865 MHz), passives	1,499	2,998
Core Meteorological	BME280, wind speed/direction pair, DS18B20, SEN0575, LTR-390	4,960	9,920
Particulate & Gases	PMS5003, SGP41, MICS-6814, SCD41	7,500	15,000
Specialized Sensing	AS3935 (CJMCU-3935)	1,800	3,600
Power Architecture	18650 cell, 5W solar panel, CN3065, holder/wiring	1,370	2,740
Fabrication	PCB (5x batch), PETG, conformal coating	2,000	4,000
Subtotal		**19,129**	**38,258**
Shared NRE & contingency	Calibration jigs, spare sensors, shipping, testing	—	11,742
Total	**Complete research-grade node(s)**	**19,129**	**50,000**

while reserving fully equipped INR 19k nodes for strategic choke points in the valley.

V. Conclusion

The First Prototype Stage of the Edge AI Portable Weather Station represents a feasible, economically disruptive approach to microclimate monitoring. By pairing the machine-learning acceleration of the ESP32-S3 with the offline resilience of an 865 MHz LoRa mesh network, the system overcomes the geographic and infrastructural limitations of centralized weather tracking. The complete firmware implementation, including sensor drivers, XGBoost inference engine, and mesh communication stack, has been developed and is ready for hardware integration. The immediate next phase involves finalizing hardware integration of the 12-sensor array across both nodes, capturing baseline tabular data, deploying the C++-converted boosting algorithm, and validating power consumption and LoRa mesh handoff between the two nodes during localized inference loops. Once validated, the node pair is ready to replicate and scale across mountainous topography, establishing the foundation of a true swarm-intelligence environmental network.

Acknowledgment

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